

Education	Computer Science BA — University of Cambridge, Trinity College (2021–24) Full Merit Scholarship; Second Highest Scoring Dissertation; Strong 2:1 overall Advanced Mathematics Class — Fazekas Mihály Grammar School (2015–21) A*A*A* equivalent final grades, Medallion of Excellence for outstanding performance Artificial Intelligence Technology Training — KÜRT Academy (2018–19) Completed as the youngest trainee ever (age 16); Explored a wide range of ML techniques	Bence Hervay Quantitative Research Engineer ✉ bh525@cantab.ac.uk ☎ +44 74 555 76542 📍 London Cambridge
	Skills Proficient in Python and C++ with 200,000+ lines experience; Quick to master new frameworks and technologies Quantitative and statistical methods, algorithms and data structures, machine learning, low-level optimisation Combining analytical thinking with cross-disciplinary STEM expertise, approaching theoretical and practical problems from first principles and producing innovative, evidence-based, data-driven solutions	
	Honours University of Cambridge — Second Best Computer Science Dissertation (2024) in the cohort of 120+ students Trinity College, Cambridge — Full Scholarship (2021) awarded to only 5 out of 200+ for academic excellence National Olympiads — 1st in Physics (2020); 1st in Mathematics (2018); Consistently national top 10 (2017–21) FIRST Global Challenge — Robotics World Champions (2017); 150+ countries; invited to the White House Over 25 national top 3 placements across Mathematics, Computer Science, Physics, and Robotics (2016–21)	
Work Experience	Quantitative Research Engineer — AGITProp (2024–present) Automatic detection and statistical refinement of financially interpretable factors for covariance matrix prediction Primary ownership of numerous key modules in the pipeline (mainly mathematical components) Key role in project planning and coordination , contributing to firm-level prioritisation decisions Constant expansion and maintenance of the infrastructure (e.g. quantitative modules, data pipeline, testing)	
	Software Developer Intern — DRW (Jun–Sep 2023) Created a fully automatic monitoring and resource allocation tool interacting with 200+ employees daily Utilised NLP techniques to filter and classify bug reports mentioned in messages for efficiency	
	Quantitative Research Intern — WorldQuant (Jun–Aug 2022) Generalised index rebalance alphas by automatic inference of rebalance schedules and methodologies Developed a framework to mass-produce low-correlation alphas using decision forests	
Projects & Research	Senior Group Leader — Alphacademy (2021–23) Member of the founding team having interviewed, onboarded, and supervised numerous other teachers Taught programming to 100+ students from novice to Olympiad level	
	Rupert's Problem — C++ & TypeScript (2024–present) Solved the central open problem of a 350-year-old mathematical topic: whether every convex polyhedron fits through itself Heavily computer-assisted proof involving 10,000+ lines of C++ utilising interval arithmetic to bound floating-point rounding errors Novel solutions to numerous complex sub-problems, including fuzzy geometric convex hull and elliptical intersection algorithms Largest independent research project with 1,000+ hours invested; comprehensive preprint in preparation.	
	Reinforcement Learning for 2048 — Cambridge Dissertation — C++ (2023–24) Developed the world's highest-scoring algorithm and the fastest simulation engine for the 2048 Puzzle Game Reaches the 2048-tile in 1.2 milliseconds with a 99.8% success rate and 75% of the time even the 32768-tile Trained a custom 200M-parameter RAMnet via self-play reinforcement learning on 150M games Combined expectimax with locally brute-forced critical phases to maximise the probability of survival	
	3D Path Tracer — Java (2021–22) Simulated reflection, refraction, texture mapping, soft shadows, and anti-aliasing for a family of implicit surfaces Won the Cambridge rendering competition held across the entire cohort of 120+ computer science students	
	Chess AI — Java (2020–21) Built a 2200+ Elo chess engine with negamax search, alpha-beta pruning, iterative deepening, transposition tables, opening books Advanced heuristic incorporating pawn structure, king safety, and piece-square tables; fine-tuned with optimisation through self-play Awarded a Full Research Scholarship by the National Research Excellence Program	
	Project Euler — Python & C++ (2019–present) Collection of challenges requiring a combination of mathematical research, advanced algorithm design, and optimisation Top 0.04% of all users by solving ~250 problems in total, including each of the first 222	
	Muon Detector — Wigner Particle Physics Research Centre (2018–19) Built a fully functional muon tomography device (stacked multi-wire proportional chamber) to map directional atmospheric muon flux Invited to the Large Hadron Collider (and visited the CMS detector) to present our findings	
	Selected Additional Projects (2015–present) Crafted entirely from scratch and without relying on high-level libraries to ensure deep understanding of the fundamentals Automatic differentiation framework in C++; Handwritten digit classifier DNN with a GUI; Physics engine; Rubik's cube solver with a GUI; Time-travel based platformer game; Solvers for Multiplayer Pac-Man, Sudoku, Nonogram, and numerous other strategic games	
	Hobbies & Sports Rubik's Cube speedsolving — 14-second average; 20x national top 20 Ultimate Frisbee — UK National Championship 6th (2023) Great Court Run — 3rd (2021); 2nd (2022); 1st and Beat the Clock (2023) Acrobatic Basketball World Cup 4th (2018) — FaceTeam Youngsters	Activities Regular instructor at the European Summer Program on Rationality Events Officer of Cambridge University Hungarian Society Captain of Trinity College Ultimate Frisbee Society Always happy to discuss interesting puzzles and brain teasers